

ROMA TRE · A.Y. 2026/27

# Policy Evaluation Lab

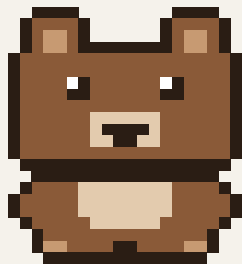
*Lecture 5: Social experiments (II) – the RAND HIE*

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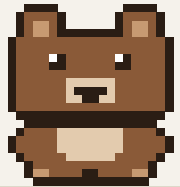
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PART 1

# The RAND experiment

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# RAND Health Insurance Experiment (I)

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One of the most influential experiments in the history of research.

- The Health Insurance Experiment (HIE) ran from 1974 to 1982 and involved about 4,000 people aged 14 to 61, from six areas of the US
- It excluded people on Medicare, Medicaid and military health insurance
- Participants were randomly assigned to one of 14 different health insurance (HI) plans:
  - no premiums for HI, but different levels of **cost-sharing**
  - enrolled in a plan for 3 or 5 years, giving up any previous insurance
  - and paid a monthly fee unrelated to the use of health care



# RAND Health Insurance Experiment (II)

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Research questions:

- What is the **price elasticity** of health-care demand?
  - whether, and by how much, health-care use falls when its price increases
- Does more generous HI lead to **better health outcomes**?



# Issues

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- Too many HI plans → treatment groups too small for comparisons to be statistically precise
- Solution: grouping similar HI plans together, from 14 to 4
  - Two extremes: comprehensive **free** care, and a **catastrophic** plan with 95% of health costs to pay
  - Two intermediate plans: **coinsurance** (covers a percentage of charges, with an income cap) and **deductible** (participants pay outpatient charges, with limits)



PART 2

# Balance check

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# Balance check

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- First step: **checking for balance**. Do subjects randomly assigned to the treatment and control groups look similar?
- compare **pre-treatment** demographic characteristics and health data
- We expect only small differences
- In Panel A of the next table, standard errors of the differences are in parentheses (cols. 2–5), and standard deviations in brackets (col. 1)
- Reference category: the catastrophic plan. The averages of every other plan are compared with the average in col. 1, and the differences are reported



# Comparison of (pre-treatment) individual characteristics

|                                       | Means                    | Differences between plan groups  |                                   |                            |                                     |
|---------------------------------------|--------------------------|----------------------------------|-----------------------------------|----------------------------|-------------------------------------|
|                                       | Catastrophic plan<br>(1) | Deductible – catastrophic<br>(2) | Coinsurance – catastrophic<br>(3) | Free – catastrophic<br>(4) | Any insurance – catastrophic<br>(5) |
| <i>A. Demographic characteristics</i> |                          |                                  |                                   |                            |                                     |
| Female                                | .560                     | -.023<br>(.016)                  | -.025<br>(.015)                   | -.038<br>(.015)            | -.030<br>(.013)                     |
| Nonwhite                              | .172                     | -.019<br>(.027)                  | -.027<br>(.025)                   | -.028<br>(.025)            | -.025<br>(.022)                     |
| Age                                   | 32.4<br>[12.9]           | .56<br>(.68)                     | .97<br>(.65)                      | .43<br>(.61)               | .64<br>(.54)                        |
| Education                             | 12.1<br>[2.9]            | -.16<br>(.19)                    | -.06<br>(.19)                     | -.26<br>(.18)              | -.17<br>(.16)                       |
| Family income                         | 31,603<br>[18,148]       | -2,104<br>(1,384)                | 970<br>(1,389)                    | -976<br>(1,345)            | -654<br>(1,181)                     |
| Hospitalized last year                | .115                     | .004<br>(.016)                   | -.002<br>(.015)                   | .001<br>(.015)             | .001<br>(.013)                      |



## Results of the balance check: individual characteristics

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- Reasonably good balance
- Differences are mostly small; only one difference is statistically significant
  - Women are less likely to be in the free insurance group
- However, when we make many comparisons, a few significant differences are expected purely by chance
  - we are looking at one of many possible draws from the sampled population
  - This pattern clearly differs from the NHIS data in Table 2 (Lecture 2)
  - Furthermore, the standard errors are relatively small, indicating more precise estimates



# Comparison of (pre-treatment) individual health

|                                     | Means                    | Differences between plan groups  |                                   |                            |                                     |
|-------------------------------------|--------------------------|----------------------------------|-----------------------------------|----------------------------|-------------------------------------|
|                                     | Catastrophic plan<br>(1) | Deductible – catastrophic<br>(2) | Coinsurance – catastrophic<br>(3) | Free – catastrophic<br>(4) | Any insurance – catastrophic<br>(5) |
| <i>B. Baseline health variables</i> |                          |                                  |                                   |                            |                                     |
| General health index                | 70.9<br>[14.9]           | –1.44<br>(.95)                   | .21<br>(.92)                      | –1.31<br>(.87)             | –.93<br>(.77)                       |
| Cholesterol (mg/dl)                 | 207<br>[40]              | –1.42<br>(2.99)                  | –1.93<br>(2.76)                   | –5.25<br>(2.70)            | –3.19<br>(2.29)                     |
| Systolic blood pressure (mm Hg)     | 122<br>[17]              | 2.32<br>(1.15)                   | .91<br>(1.08)                     | 1.12<br>(1.01)             | 1.39<br>(.90)                       |
| Mental health index                 | 73.8<br>[14.3]           | –.12<br>(.82)                    | 1.19<br>(.81)                     | .89<br>(.77)               | .71<br>(.68)                        |
| Number enrolled                     | 759                      | 881                              | 1,022                             | 1,295                      | 3,198                               |



## Results of the balance check: individual health

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- No statistically significant differences in the general health index
- Other differences, in cholesterol, blood pressure and mental health, are also unrelated to treatment assignment
  - Lower cholesterol (better health) in the free insurance group, but
  - the same group has a lower general health index (poorer health)
  - The lack of a consistent pattern shows that these gaps are due to chance

→ **No systematic differences:** randomisation produced balanced groups



PART 3

# Results

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## RAND HIE results: health-care use

|                           | Means                    | Differences between plan groups  |                                   |                            |                                     |
|---------------------------|--------------------------|----------------------------------|-----------------------------------|----------------------------|-------------------------------------|
|                           | Catastrophic plan<br>(1) | Deductible – catastrophic<br>(2) | Coinsurance – catastrophic<br>(3) | Free – catastrophic<br>(4) | Any insurance – catastrophic<br>(5) |
| <i>A. Health-care use</i> |                          |                                  |                                   |                            |                                     |
| Face-to-face visits       | 2.78<br>[5.50]           | .19<br>(.25)                     | .48<br>(.24)                      | 1.66<br>(.25)              | .90<br>(.20)                        |
| Outpatient expenses       | 248<br>[488]             | 42<br>(21)                       | 60<br>(21)                        | 169<br>(20)                | 101<br>(17)                         |
| Hospital admissions       | .099<br>[.379]           | .016<br>(.011)                   | .002<br>(.011)                    | .029<br>(.010)             | .017<br>(.009)                      |
| Inpatient expenses        | 388<br>[2,308]           | 72<br>(69)                       | 93<br>(73)                        | 116<br>(60)                | 97<br>(53)                          |
| Total expenses            | 636<br>[2,535]           | 114<br>(79)                      | 152<br>(85)                       | 285<br>(72)                | 198<br>(63)                         |



## RAND HIE results: health-care use

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- Individuals with more generous HI used more health care
  - lower prices, higher demand
- Hospital admissions are less sensitive to price than outpatient care
- In the free plan, outpatient expenses increase by about two-thirds (169/248), and total expenses by 45% (285/636), compared with the catastrophic plan
- Most differences, in particular those for the free plan, are statistically significant

The **difference-in-means** estimator is unbiased when individuals are randomly assigned to treatment



## RAND HIE results: health outcomes

|                                 | Means                    | Differences between plan groups  |                                   |                            |                                     |
|---------------------------------|--------------------------|----------------------------------|-----------------------------------|----------------------------|-------------------------------------|
|                                 | Catastrophic plan<br>(1) | Deductible – catastrophic<br>(2) | Coinsurance – catastrophic<br>(3) | Free – catastrophic<br>(4) | Any insurance – catastrophic<br>(5) |
| <i>B. Health outcomes</i>       |                          |                                  |                                   |                            |                                     |
| General health index            | 68.5<br>[15.9]           | -.87<br>(.96)                    | .61<br>(.90)                      | -.78<br>(.87)              | -.36<br>(.77)                       |
| Cholesterol (mg/dl)             | 203<br>[42]              | .69<br>(2.57)                    | -2.31<br>(2.47)                   | -1.83<br>(2.39)            | -1.32<br>(2.08)                     |
| Systolic blood pressure (mm Hg) | 122<br>[19]              | 1.17<br>(1.06)                   | -1.39<br>(.99)                    | -.52<br>(.93)              | -.36<br>(.85)                       |
| Mental health index             | 75.5<br>[14.8]           | .45<br>(.91)                     | 1.07<br>(.87)                     | .43<br>(.83)               | .64<br>(.75)                        |
| Number enrolled                 | 759                      | 881                              | 1,022                             | 1,295                      | 3,198                               |

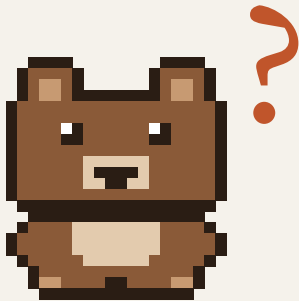


## RAND HIE results: conclusions

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- However, there are no statistically significant differences in health outcomes between the groups
- All indicators are similar across groups
- Evidence that generous health insurance can increase costs without promoting better health
- The RAND experiment had a few drawbacks:
  - All groups had at least some HI coverage
  - **External validity** was a concern: today's uninsured Americans are in many ways different from the HIE participants





# Questions?

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